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## PROCESSING WRITE OPERATION INFORMATION USING INFLATED DATA

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### Abstract

A technique that processes write operation information involves receiving, as the write operation information, user data and corresponding metadata. The technique further involves generating inflated data which includes the user data and the corresponding metadata. The inflated data has a size that is larger than a total (or overall) size of the user data and the corresponding metadata. Additionally, the inflated data naturally aligns on storage boundaries of solid state device (SSD) storage (non-volatile flash memory). The technique further involves, after generating the inflated data, storing the inflated data in the SSD storage.

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### Background/Summary

## BACKGROUND

[0001] A conventional data storage system includes an input buffer which temporarily holds write data en route to secondary storage. The input buffer is formed of high performance and high endurance memory such as SRAM (static random access memory) which behaves as non-volatile memory by using a battery to maintain a constant charge. The secondary storage is formed of slower, less expensive memory such as solid state device (SSD) memory (non-volatile flash memory), hard disk drive (HDD) memory (magnetic disk drive memory), combinations thereof, etc.

[0002] During operation, the conventional data storage system may acknowledge completion of write IOs (write input/output operations) as soon as the write data is stored in the input buffer. Such write-back operation provides faster response times (i.e., lower latency) than write-thru operation in which acknowledgement is provided only after the write data is bound (or committed) to the secondary storage.

## SUMMARY

[0003] Unfortunately, there are deficiencies to conventional data storage systems which use high performance and high endurance memory such as SRAM as an input buffer. Along these lines, SRAM requires additional resources such as a battery to provide backup power to prevent data loss in the event of a power outage. Additionally, such memory, as well as high end NVRAM, is relatively scarce thus imposing restrictions/limitations on input buffer size as well as increasing the overall cost of the data storage systems. Furthermore, such memory tends to see frequent technology changes and thus has limited lifetime (e.g., shortened life cycles to stay with the best technology and maintain minimum possible latency), typically imposes different support and/or maintenance requirements based on the specific memory type/vendor/etc., and so on.

[0004] Improved techniques are directed to processing write operation information using inflated data which naturally aligns on storage boundaries of SSD storage (e.g., regular NAND flash memory). Such techniques enable writing user data and corresponding metadata as full writes which are aligned with the native page size of the SSD storage. Accordingly, such operation alleviates the need to perform read-modify-writes that would otherwise increase latency (e.g., there is no need to read data from the SSD storage, modify the read data to fit in new data with smaller granularity, and then write the modified data back to the SSD storage). Moreover, such operation enables the use of SSD storage as a write cache (or similar logfile or journal) thus lowering costs, removing the need for additional resources such as a battery to persist data, and alleviating the need to deal with frequent technology changes in higher performance and higher endurance memory.

[0005] One embodiment is directed to a method of processing write operation information. The method includes receiving, as the write operation information, user data and corresponding metadata. The method further includes generating inflated data which includes the user data and the corresponding metadata. The inflated data has a size that is larger than a total size of the user data and the corresponding metadata. Additionally, the inflated data naturally aligns on storage boundaries of SSD storage. The method further includes, after generating the inflated data, storing the inflated data in the SSD storage.

[0006] Another embodiment is directed to storage equipment which includes SSD storage, and control circuitry coupled with the SSD storage. The control circuitry is constructed and arranged to perform a method of: [0007] (A) receiving, as write operation information, user data and corresponding metadata, [0008] (B) generating inflated data which includes the user data and the corresponding metadata, the inflated data having a size that is larger than a total size of the user data and the corresponding metadata, and the inflated data naturally aligning on storage boundaries of the SSD storage, and [0009] (C) after generating the inflated data, storing the inflated data in the SSD storage.

[0010] Yet another embodiment is directed to a computer program product having a non-transitory

computer readable medium which stores a set of instructions to process write operation information. The set of instructions, when carried out by computerized circuitry, causes the computerized circuitry to perform a method of: [0011] (A) receiving, as the write operation information, user data and corresponding metadata; [0012] (B) generating inflated data which includes the user data and the corresponding metadata, the inflated data having a size that is larger than a total size of the user data and the corresponding metadata, and the inflated data naturally aligning on storage boundaries of SSD storage; and [0013] (C) after generating the inflated data, storing the inflated data in the SSD storage.

[0014] In some arrangements, the SSD storage forms a write cache of a storage array. Additionally, storing the inflated data in the SSD storage includes writing the inflated data into the write cache of the storage array.

[0015] In some arrangements, the user data is application data. Additionally, the corresponding metadata is constructed and arranged to identify write operation details for storing the application data in non-volatile memory of the storage array, the non-volatile memory being different from the write cache. Furthermore, the corresponding metadata has a size which is smaller than that of the application data. Also, generating the inflated data includes padding the corresponding metadata to form padded metadata having an inflated size that is larger than the size of the corresponding metadata.

[0016] In some arrangements, writing the inflated data into the write cache includes writing, as the inflated data, the application data and the padded metadata into the write cache to enable late binding of the application data into the non-volatile memory.

[0017] In some arrangements, the method further includes, after the application data and the padded metadata are written into the write cache, performing a late bind operation which commits the application data to the non-volatile memory of the storage array.

[0018] In some arrangements, the method further includes, prior to receiving the user data and the corresponding metadata, performing a comparison operation which compares a size of the user data to a threshold and provides a comparison result indicating that the size of the user data is greater than the threshold.

[0019] In some arrangements, padding the corresponding metadata includes, in response to the comparison result indicating that the size of the user data is greater than the threshold, creating a full page which includes the corresponding metadata and zeroes, the full page being aligned with a native page size of the SSD storage.

[0020] In some arrangements, the method further includes: [0021] (i) receiving additional user data and corresponding additional metadata; [0022] (ii) performing an additional comparison operation which compares a size of the additional user data to the threshold and provides an additional comparison result indicating that the size of the additional user data is less than the threshold; and [0023] (iii) in response to the additional comparison result indicating that the size of the user data is less than the threshold, refraining from generating additional inflated data which includes the additional user data and the corresponding additional metadata.

[0024] In some arrangements, the method further includes, prior to receiving the user data and the corresponding metadata, forming the write cache from a set of SSDs.

[0025] In some arrangements, the set of SSDs provides X amount of storage, X amount being at least 1 Terabyte. Additionally, forming the write cache from the set of SSDs includes: [0026] (i) allocating 10% of the X amount of storage to the write cache, and [0027] (ii) allocating at least 80% of the X amount of storage to a garbage collection service to minimize write amplification resulting from garbage collecting the write cache.

[0028] It should be understood that, in the cloud context, at least some electronic circuitry is formed by remote computer resources distributed over a network (e.g., hosts, management equipment, scalable storage resources, etc.). Such an electronic environment is capable of providing certain advantages such as high availability and data protection, transparent operation

and enhanced security, big data analysis, etc.

[0029] Other embodiments are directed to electronic systems and apparatus, processing circuits, computer program products, and so on. Some embodiments are directed to various methods, electronic assemblies, components and circuitry which are involved in processing write operation information using inflated data.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0030] The foregoing and other objects, features and advantages will be apparent from the following description of particular embodiments of the present disclosure, as illustrated in the accompanying drawings in which like reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating the principles of various embodiments of the present disclosure.

[0031] FIG. 1 is a block diagram of an electronic setting which processes write operation information using inflated data in accordance with certain embodiments in accordance with certain embodiments.

[0032] FIG. 2 is a block diagram of electronic circuitry (or equipment) suitable for the electronic setting in accordance with certain embodiments.

[0033] FIG. 3 is a block diagram of example incoming data processing in accordance with embodiments.

[0034] FIG. 4 is a block diagram of other example incoming data processing in accordance with embodiments.

[0035] FIG. 5 is a block diagram of additional example incoming data processing in accordance with embodiments.

[0036] FIG. 6 is a flowchart of a procedure to process write operation information using inflated data in accordance with certain embodiments.

### DETAILED DESCRIPTION

[0037] An improved technique is directed to processing write operation information using inflated data which naturally aligns on storage boundaries of solid state device (SSD) storage (e.g., routine NAND flash memory or similar commodity hardware). Such a technique enables writing user data and corresponding metadata as full writes which are aligned with the native page size of the SSD storage. Accordingly, such operation alleviates the need to perform read-modify-writes that would otherwise increase latency (e.g., there is no need to read data from the SSD storage, modify the read data to fit in new data with smaller granularity, and write the modified data back to the SSD storage). Moreover, such operation enables the use of regular SSD storage as a write cache, logfile, journal, etc., thus lowering costs, removing the need for additional resources such as batteries to persist data, and alleviating the need to deal with frequent technology changes in higher performance and higher endurance memory.

[0038] FIG. 1 shows an electronic setting **100** in which storage equipment processes write operation information using inflated data in accordance with certain embodiments. By way of example, the electronic setting **100** is a data storage environment that includes host computers **102(1)**, **102(2)**, . . . (collectively, host computers **102**), data storage equipment **104**, a communications medium **106**, and perhaps other equipment **108**.

[0039] It should be appreciated that a variety of electronic situations are suitable for use. Along these lines, such situations may take the form of general purpose computing environments, server farms, cloud-based systems, enterprise/corporate computing/infrastructure equipment, retail settings, manufacturing settings, government settings, specialized service providers, academic settings, websites, environments for specialized operations, combinations thereof, and so on.

[0040] Each host computer **102** is constructed and arranged to perform useful work. For example, one or more of the host computers **102** may operate as a file server, a web server, an email server, an enterprise server, a database server, a transaction server, combinations thereof, etc. which provides host input/output (I/O) requests **120** to the data storage equipment **104**. In this context, the host computers **102** may provide a variety of different I/O requests **120** (e.g., block and/or file based write commands, block and/or file based read commands, combinations thereof, etc.) that direct the data storage equipment **104** to store host data **122** within and retrieve host data **122** from storage (e.g., primary storage or main memory, secondary storage, tiered storage, combinations thereof, etc.).

[0041] The data storage equipment **104** is an example of electronic equipment that is capable of processing write operation information using inflated data. The data storage equipment **104** includes an electronic equipment infrastructure **140**, storage processing circuitry **142**, a set of storage devices **144**, and other devices **146**.

[0042] The electronic equipment infrastructure **140** is constructed and arranged to support operation of various components of the data storage equipment **104**. To this end, the electronic equipment infrastructure **140** positions the various components relative to each other, conveys power signals from power converters to the various components, positions fans along air pathways to provide cooling to the various components (i.e., to remove heat), protects the various components against tampering, damage, and so on. Along these lines, the electronic equipment infrastructure **140** may include one or more frames or card cages, chassis/housings/cabinets, power supplies/converters, fans, backplanes or midplanes, and so on.

[0043] The storage processing circuitry **142** is constructed and arranged to respond to the host I/O requests **120** received from the host computers **102** by writing data into the set of storage devices **144** and reading the data from the set of storage devices **144**. The storage processing circuitry **142** may include one or more physical storage processing modules or engines, data movers, director boards, blades, etc. In accordance with certain embodiments, the storage processing circuitry **142** may include a variety of specialized subcomponents such as processing circuitry to process I/O requests **120** from the host computers **102**, cache memory to operate as read and/or write caches, on-board NICs to provide communications between the storage processing circuitry **142** and/or external devices, LEDs, switches for local I/O, and so on.

[0044] The set of storage devices **144** is constructed and arranged to store data within the data storage equipment **104**. In accordance with certain embodiments, the set of storage devices **44** may arrange the data in accordance with one or more data protection schemes (e.g., RAID1, RAID5, RAID6, RAID10, etc.). Example storage devices **144** include RAM devices, NVRAM devices, other solid state memory devices (SSDs), hard disk drives (HDDs), combinations thereof, and so on.

[0045] The other devices **146** of the data storage equipment **104** are constructed and arranged to provide supporting features. Along these lines, the other devices **146** may include a set of communications interfaces (e.g., a set of network interface controllers or NICs) that provide communications for the data storage equipment **104**, a user interface for local user access, backup power sources, and so on.

[0046] The communications medium **106** is constructed and arranged to connect the various components of the electronic setting **100** together to enable these components to exchange electronic signals **150** (e.g., see the double arrow **150**). At least a portion of the communications medium **106** is illustrated as a cloud to indicate that the communications medium **106** is capable of having a variety of different topologies including backbone, hub-and-spoke, loop, irregular, combinations thereof, and so on. Along these lines, the communications medium **106** may include copper-based data communications devices and cabling, fiber optic devices and cabling, wireless devices, combinations thereof, etc. Furthermore, the communications medium **108** is capable of supporting LAN-based communications, SAN-based communications, cellular communications,

WAN-based communications, distributed infrastructure communications, other topologies, combinations thereof, etc.

[0047] The other equipment **108** represents other possible componentry of the electronic setting **100**. Along these lines, the other equipment **108** may include remote data storage equipment that provides data to and/or receives data from the data storage equipment **104** (e.g., replication arrays, backup and/or archiving equipment, external service processors and/or other management/control devices, etc.).

[0048] During operation, the data storage equipment **104** processes I/O requests **120** from the set of host computers **102** to perform useful work. In particular, the storage processing circuitry **142** writes host data **122** into and retrieves host data **122** from the set of storage devices **144** in response to the I/O requests **120**.

[0049] During such operation, the storage processing circuitry **142** is capable of processing write operation information using inflated data which naturally aligns on storage boundaries of SSD storage. In some arrangements, the storage processing circuitry **142** inflates the data when the data is smaller than a predefined size threshold and does not inflate the data when the data is larger than the predefined size threshold in order to prevent excessively consuming bandwidth (e.g., bandwidth saturation).

[0050] Such data inflation alleviates the need to perform read-modify-writes when writing data to the SSD storage that would otherwise increase latency (e.g., there is no need to read data from the SSD storage, modify the read data to fit in the new data with smaller granularity, and then write the modified data to the SSD storage). Moreover, such data inflation enables the use of SSD storage as a write cache thus lowering costs, removing the need for additional resources such as a battery to persist data, and alleviating the need to deal with frequent technology changes, trends, etc. in higher performance and higher endurance memory. Further details will now be provided with reference to FIG. 2.

[0051] FIG. 2 shows electronic circuitry **200** which is suitable for at least a portion of the data storage equipment **104** of the electronic setting **100** in accordance with certain embodiments (also see the storage processing circuitry **142** in FIG. 1). The electronic circuitry **200** includes a set of interfaces **202**, memory **204**, processing circuitry **206**, and other componentry (or circuitry) **208**.

[0052] The set of interfaces **202** is constructed and arranged to connect the electronic circuitry **200** to the communications medium **106** (FIG. 1) to enable communications with other devices of the electronic setting **100**. Such communications may be IP-based, SAN-based, cellular-based, cable-based, fiber-optic based, wireless, cloud-based, combinations thereof, and so on. Additionally, the set of interfaces **202** is constructed and arranged to connect the electronic circuitry **200** to other local equipment. Accordingly, the set of interfaces **202** may include one or more host interfaces (e.g., a computer network interface, a fibre-channel interface, etc.), one or more storage device interfaces (e.g., a host adapter or HBA, etc.), and other interfaces. As a result, the set of interfaces **202** enables the electronic circuitry **200** to robustly and reliably communicate with various apparatus.

[0053] The memory **204** is intended to represent both volatile storage (e.g., DRAM, SRAM, etc.) and non-volatile storage (e.g., solid state memory, magnetic memory, etc.). The memory **204** stores a variety of software constructs **220** including an operating system **222**, specialized code **224**, and other code and data **226**. The operating system **222** refers to particular control code such as a kernel to manage computerized resources (e.g., processor cycles, memory space, etc.), the I/O stack (e.g., drivers), and so on. The specialized code **224** refers to particular instructions for processing write operation information using inflated data which naturally aligns on storage boundaries of SSD storage. The other code and data **226** refers to control parameters/thresholds, tools, routines, test procedures, statistics, combinations thereof, and so on.

[0054] The processing circuitry **206** is constructed and arranged to operate in accordance with the various software constructs **220** stored in the memory **204**. Along these lines, the processing

circuitry **206** may execute the specialized code **224** to form specialized circuitry that robustly and reliably enables the electronic circuitry **200** to process write operation information using inflated data which naturally aligns on storage boundaries of SSD storage. Such processing circuitry **206** may be implemented in a variety of ways including via one or more processors (or cores) running specialized software, application specific ICs (ASICs), field programmable gate arrays (FPGAs) and associated programs, discrete components, analog circuits, other hardware circuitry, combinations thereof, and so on. In the context of one or more processors executing software, a computer program product **240** is capable of delivering all or portions of the software constructs **220** to the electronic circuitry **200**. In particular, the computer program product **240** has a non-transitory (or non-volatile) computer readable medium which stores a set of instructions that controls one or more operations of the electronic circuitry **200**. Examples of suitable computer readable storage media include tangible articles of manufacture and apparatus which store instructions in a non-volatile manner such as DVD, CD-ROM, flash memory, disk memory, tape memory, and the like.

[0055] The other componentry **208** refers to other hardware of the electronic circuitry **200**. Along these lines, the electronic circuitry **200** may further include specialized equipment such as a local user I/O circuitry, circuitry for fault tolerance, etc. Further details will now be provided with reference to FIG. 3.

[0056] FIG. 3 shows a view **300** illustrating how the electronic circuitry **200** (FIG. 2) may process example data using inflated data in accordance with certain embodiments. The view **300** includes a write cache **310** and secondary storage **320** (also see the data storage equipment **104** in FIG. 1 and the electronic circuitry **200** in FIG. 2).

[0057] The write cache **310** serves as temporary storage for write operation information for a write request. Such a write request may originate from a variety of sources such as a host (e.g., see the host request **120** from one of the host computers **102** in FIG. 1), a peer storage processor, another node of a cluster, and so on. Moreover, the write cache **310** may be constructed and arranged to store the write operation information in the form a logfile, a journal, pages of virtual memory, combinations thereof, and so on.

[0058] Use of such a write cache **310** enables the write request to be acknowledged as soon as the write operation information is safely stored in the write cache **310**. Such write-back processing reduces latency since the source of the write request may receive acknowledgement prior to data contained within the write request actually being committed to (via late binding) the secondary storage **320**.

[0059] In accordance with certain embodiments, the write cache **310** is formed of SSD storage or similar commodity hardware. In the context of non-volatile flash memory, the write cache **310** does not require complex/costly batteries to maintain data and/or frequent upgrading/modernization to avoid becoming obsolete.

[0060] The secondary storage **320** may be formed by various non-volatile forms of memory such as SSDs, hard disk drives (HDDs), combinations thereof, etc. In some arrangements, the secondary storage **320** may be configured into tiers (i.e., tiered storage).

[0061] By way of example, suppose that the SSD storage that forms the write cache **310** has a native page size of 4K. In such arrangements, the electronic circuitry **200** is able to write data at full page granularity at relatively fast speed (e.g., 900K IOPS). However, the electronic circuitry **200** may nevertheless be able to write data at smaller granularity to the write cache **310** such as in sizes of 512B by performing read-modify-write operations at a slower speed (e.g., at less than 200K IOPS).

[0062] At this point, it should be appreciated that minimizing latency may be achieved by writing data into the write cache **310** at the native page size (e.g., as full 4K pages). In particular, data which is less than a full page may be inflated (e.g., padded with zeroes) to align that data with the native page size of the write cache **310**. In this example, all 512B sectors may be inflated to 4 KB.

Such operation alleviates the need to perform suboptimal read-modify-write operations.

[0063] Along these lines and as shown in FIG. 3, suppose that incoming data **330** includes 4K of user data (identified with thin cross hatching) and **512B** of corresponding metadata (identified with darker cross hatching). For example, the **4K** of user data may be provided by a user level application (e.g., a client application running on a host). Additionally, the **512B** of corresponding metadata may include particular information for the write operation (e.g., a page descriptor, etc.).

[0064] Now, if the electronic circuitry **200** were to simply write the incoming data **330** to the write cache **310** as is, the electronic circuitry **200** would be able to write the **4K** of user data to the write cache **310** as a full page at a relatively fast speed. However, the electronic circuitry **200** would have to write the **512B** of corresponding metadata to the write cache **310** via a read-modify-write operation which could significantly increase the processing time (e.g., increase latency by 75%).

[0065] To avoid performing the read-modify-write operation, the electronic circuitry **200** inflates one or more portions of the incoming data **330**. In this example, the electronic circuitry **200** does not inflate the user data since the user data is already aligned with native page boundaries of the SSD storage which forms the write cache **310**. However, the electronic circuitry **200** inflates the corresponding metadata since the corresponding metadata is not currently aligned with native page boundaries of the SSD storage which forms the write cache **310**.

[0066] That is, the electronic circuitry **200** performs an inflate operation **340** which inflates the incoming data **330** to align with the native memory boundaries of the write cache **310**. Specifically, to inflate initial data of the size of 512B, the electronic circuitry **200** pads the initial data with zeroes to form inflated data in accordance with certain embodiments. Accordingly, the electronic circuitry **200** creates, from initial metadata which is 512B in size, inflated metadata which is 4K in size (e.g., by adding 3,584B of zeroes) as shown in FIG. 3. As a result, the inflated metadata is aligned with native page boundaries of the SSD storage which forms the write cache **310**.

[0067] As further shown in FIG. 3, the electronic circuitry **200** performs a caching operation **350** which stores the inflated incoming data which includes 4K of user data and now 4K of corresponding metadata (i.e., the inflated metadata) in the write cache **310**. Since the caching operation **350** involves writing only full pages, the caching operation **350** is performed very quickly (e.g., at 900K IOPS). Accordingly, for a write-back caching scheme, the electronic circuitry **200** is able to acknowledge the write request with very low latency.

[0068] Eventually, the electronic circuitry **200** performs a commit (or late binding) operation **360** which commits the user data to the secondary storage **320**. Along these lines, the electronic circuitry **200** identifies where the user data should be stored (e.g., based on the inflated metadata) and then formally synchronizes the user data in the write cache **310** with that in the secondary storage **320**.

[0069] It should be appreciated that the commit operation **320** may be performed well after the incoming data **330** has been safely stored in the write cache **310**. Accordingly, such late binding does not impact perceived performance from the host's perspective.

[0070] It should further be appreciated that the native page size for the write cache **310** was described above as being 4 KB by way of example only. In other arrangements, the native page size is different, e.g., 2 KB, 8 KB, 16 KB, 32 KB, 64 KB, 128 KB, and so on. Further details will now be provided with reference to FIGS. 4 and 5.

[0071] FIGS. 4 and 5 show additional processing examples in accordance with certain embodiments. FIG. 4 shows an example of initial data which has a relatively small footprint and which is inflated in accordance with certain embodiments. FIG. 5 shows an example of inflated data which is evaluated for possible inflating but is not inflated to reduce SSD endurance overhead (e.g., to maximize endurance) in accordance with certain embodiments.

[0072] As shown in the view **400** of FIG. 4, suppose that the electronic circuitry **200** (FIG. 2) receives incoming data for write caching in SSD storage having a 4K native page size (e.g., also see the write cache **310** in FIG. 3). For example, an application may require 8 GB to be write



cached of which 4 GB is user data and 4 GB is corresponding metadata.

[0073] As shown in FIG. 4, the incoming data **410** requires an 8 GB footprint. Along these lines, the user data is provided in 4 KB pages (1M pages initially consuming 4 GB). Additionally, the corresponding metadata is provided in **512B** sectors (8M sectors initially consuming 4 GB).

[0074] The electronic circuitry **200** performs inflate operations **410** to inflate all of the **512B** sectors of corresponding metadata to 4 KB pages. Accordingly, the electronic circuitry **200** allocates 32 GB for the 8M sectors that were inflated resulting in a footprint of 32 GB for the inflated corresponding metadata.

[0075] As a result and as shown in FIG. 4, the inflated data derived from the incoming data **410** requires an inflated 36 GB footprint. Within the 36 GB footprint, 8 GB is for user data and 32 GB is for the corresponding metadata.

[0076] It should be appreciated that the context of FIG. 4 is a write cache for storage equipment which may be provisioned with storage drives having capacities in terabytes (e.g., 4 TB, 8 TB, 16 TB, etc.). Accordingly, the 36 GB footprint for the inflated data **430** is acceptable (e.g., extremely small compared to the available SSD capacity) and enables write caching with low latency (e.g., less than 200K IOPS).

[0077] It should be further appreciated that there may be instances in which the electronic circuitry **200** does not inflate incoming data. For example, for very large incoming data, inflating may result in hitting certain bandwidth limits (e.g., saturation of a drive's PCIe link).

[0078] To address this and in accordance with certain embodiments, the electronic circuitry **200** initially compares the size of the incoming data to a predefined threshold to determine whether to perform an inflate operation. If the size of the incoming data is less than the predefined threshold, the electronic circuitry **200** inflates the incoming data to align with the native page size boundaries of the write cache and thus provide acknowledgement with low latency. However, if the size of the incoming data is greater than the predefined threshold, the electronic circuitry **200** does not inflate the incoming data thus potentially avoiding bandwidth saturation.

[0079] FIG. 5 shows a view **500** regarding certain details for the electronic circuitry **200** in accordance with certain embodiments. In particular, the view **500** includes the write cache **310** (SSD storage or similar commodity storage hardware) and a predefined threshold **510**. By way of example, the native page size of the write cache is 4 KB and the threshold **510** is 32 KB.

[0080] In the view **500**, the electronic circuitry **200** processes two example sets of incoming data **520**, **522**. The first example incoming data **520** is relatively small (e.g., less than 8 KB). However, the second example incoming data **522** is relatively large (e.g., greater than 32 KB).

[0081] For the first example incoming data **520**, the electronic circuitry **200** performs a threshold compare operation **530** in which the electronic circuitry **200** compares the size of the first example incoming data **520** to the predefined threshold **510**. By way of example, the incoming data **520** includes 4 KB of user data and 512B of corresponding metadata. A result of the threshold compare operation **530** indicates that the size of the first example incoming data **520** is lower than the threshold **510**.

[0082] In response to the comparison result, the electronic circuitry **200** performs an inflate operation **540** to inflate all 512B sectors to the native page size of the write cache **310** which is 4 KB. Accordingly, the electronic circuitry **200** generates inflated data **550** from the incoming data **520**. As shown in FIG. 5, the inflated data **550** is padded with zeroes and is 8 KB in size.

[0083] The electronic circuitry **200** then performs a caching operation **560** which writes the inflated data **550** into the write cache **310**. It should be appreciated that this write operation is relatively fast enables the electronic circuitry **200** to provide an acknowledgement in a write-back manner even though the incoming data **520** has not been stored in secondary storage (e.g., see the secondary storage **320** in FIG. 3). Moreover, inflation does not cause saturation of the write cache bandwidth.

[0084] For the second example incoming data **522**, the electronic circuitry **200** performs a threshold compare operation **570** in which the electronic circuitry **200** compares the size of the

second example incoming data **522** to the predefined threshold **510**. By way of example, the incoming data **522** includes 32 KB of user data and 512B of corresponding metadata. A result of the threshold compare operation **570** indicates that the size of the second example incoming data **520** exceeds the threshold **510** of 32 KB.

[0085] In response to the comparison result, the electronic circuitry **200** does not perform an inflate operation. Rather, the electronic circuitry **200** leaves the second example incoming data **520** uninflated.

[0086] The electronic circuitry **200** then performs a caching operation **580** which writes the second example incoming data **520** into the write cache **310**. It should be appreciated that such refraining from inflating the second example incoming data **520** due to the relatively large size prevents saturating write cache bandwidth (e.g., an NVMe PCIe link). Additionally, it should be appreciated that the relatively large size of the second example incoming data **520** is going to result in somewhat higher latency anyways (i.e., such higher latency is expected and thus satisfactory) so not inflating the second example incoming data **520** is an acceptable outcome.

[0087] Accordingly, use of the predefined threshold **510** enables selective inflation of incoming data based on incoming data size. Such selective inflation enables low latency when write cache processing incoming data that is small in size (by inflating the data into full pages). Additionally, such selective inflation avoids saturating write cache bandwidth when write cache processing incoming data that is large in size (by refraining from inflating the data).

[0088] In some embodiments, the predefined threshold **510** is derived through statistical evaluation. Along these lines, various data storage platforms may be evaluated to determine appropriate threshold values that balance low latency with possible bandwidth saturation.

[0089] Additionally, the various predefined thresholds to choose from may depend on the native page size of the memory that forms the write cache **310**. For example, if the native page size is 4 KB, suitable thresholds include 16 KB, 32 KB, 64 KB, 128 KB, and so on.

[0090] Furthermore, the particular value of the predefined threshold **510** for a particular data storage platform may depend on the configuration of data storage components within the platform. Moreover, such a threshold **510** may be adjusted/tuned over time based on evaluation of further statistics once the platform is deployed in order to accommodate factors such as user/host application behaviors, network traffic, daily usage trends, and so on. Further details will now be provided with reference to FIG. 6.

[0091] FIG. 6 is a flowchart of a procedure **600** to process write operation information using inflated data in accordance with certain embodiments. Such a procedure **600** is performed by specialized circuitry and is well suited for caching incoming data into commodity hardware such as SSD storage (also see the data storage equipment **104** in FIG. 1 and the electronic circuitry **200** in FIG. 2).

[0092] At **602**, the specialized circuitry receives, as the write operation information, user data and corresponding metadata. In some arrangements, the specialized circuitry confirms that the size of the write operation information is below a predefined threshold and only generates inflated data if the overall size of the user data and corresponding metadata (i.e., total size) is less than the predefined threshold.

[0093] At **604**, the specialized circuitry generates inflated data which includes the user data and the corresponding metadata. The inflated data has a size that is larger than a total size of the user data and the corresponding metadata. Additionally, the inflated data naturally aligns on storage boundaries of SSD storage.

[0094] At **606**, the specialized circuitry stores the inflated data in the SSD storage. At this point, the specialized circuitry may provide acknowledgement back to the source of the write operation information. At some point and as further indicated in **606**, the specialized circuitry writes the data to secondary storage (e.g., via late binding).

[0095] As described above, improved techniques are directed to processing write operation

information using inflated data which naturally aligns on storage boundaries of SSD storage (e.g., commodity hardware that forms a write cache **310**). Such techniques enable writing user data and corresponding metadata as full writes which are aligned with the native page size of the SSD storage. Accordingly, such operation alleviates the need to perform read-modify-write (RMW) operations that would otherwise increase latency (e.g., there is no need to read data from the SSD storage, modify the read data to fit in the new data with smaller granularity, and then write the modified data to the SSD storage). Moreover, such operation enables the use of standard SSD storage as a write cache thus removing the need for additional resources such as a battery to persist data, alleviating the need to deal with frequent technology changes in higher performance and higher endurance memory, and lowering costs.

[0096] While various embodiments of the present disclosure have been particularly shown and described, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure as defined by the appended claims.

[0097] Additionally, it should be understood that various components of the electronic setting **100** such as the host computers **102** are capable of being implemented in or “moved to” the cloud, i.e., to remote computer resources distributed over a network. Here, the various computer resources may be distributed tightly (e.g., a server farm in a single facility) or over relatively large distances (e.g., over a campus, in different cities, coast to coast, etc.). In these situations, the network connecting the resources is capable of having a variety of different topologies including backbone, hub-and-spoke, loop, irregular, combinations thereof, and so on. Additionally, the network may include copper-based data communications devices and cabling, fiber optic devices and cabling, wireless devices, combinations thereof, etc. Furthermore, the network is capable of supporting LAN-based communications, SAN-based communications, combinations thereof, and so on.

[0098] It should be appreciated that, in the era of competition of all-flash-arrays, one of the characteristics by which an array may be measured is via write latency/response-time. To achieve the best write-operation response-time possible, some arrays may use a low-latency log/journal/write-cache that stores the data and a minimal set of metadata to minimize the time-to-ack. A late bind that properly stores the data within the array's data-structure is then performed at a later phase. The data that is being written to the log/journal/write-cache must be persistent and protected in the sense that it must sustain any reboot or power-outage. Losing this data before is destaged is unacceptable and will be considered a data-loss. The devices used for log/journal/write-cache must provide low-latency and high endurance to sustain the high amount of data-written to them on a daily basis.

[0099] NVMe PCIe block devices such as NVRAMs, provide high performance and high endurance which can be leveraged by storage systems as their log/journal/write-cache. However, NVRAMs are currently not common, scarcely sourced and require a relatively large battery. This technology has a limited lifetime and may require integration of new types/vendors every few years, while these integrations are long and tedious and may create permutation of the product line.

[0100] Using common SSDs (e.g., commodity hardware) as log/journal/write-cache would be preferable but these pose the challenge of these drives being worn-out ahead of time, and the limited performance and response-time they provide out-of-the-box. Using high endurance SSDs may be too expensive resulting in a non-competitive solution.

[0101] Disclosed herein and in accordance with certain embodiments, a mechanism serves as a cost-effective alternative for storage systems which can provide similar behavior using commodity hardware. To achieve that both (a) performance, (b) endurance of such SSDs is considered.

[0102] A goal, in accordance with certain embodiments is to use a regular SSD for the log/journal/write-cache without compromising the performance and endurance while maintaining the cost-effectiveness of the solution. Such use enables a storage system persistent-cache using commodity hardware to be achieved.

[0103] It should be appreciated that it is common for an SSD to have an indirection unit (IU) of at least 4 KB, while still maintaining an external sector size of 512B which allows 512B writes. Accordingly, the storage system may write small sector size (512B) blocks as well as native 4k blocks.

[0104] In some storage arrays, metadata writes are done at 512B granularity, while user-data writes are done at 4K page size granularity, while most write operations include both user-data and corresponding metadata. In addition, user-data may also be written in sub-4K IO sizes.

[0105] Typically, the SSD 512B-write-performance (<200k IOPS) is lower than its 4 KB write performance (900k IOPS), as drives need to read-modify-write (RMW) a 4K blocks to accomplish a 512B write. In some cases, the drive will perform a synchronous operation, meaning write latency equals to read+write latency, while in other cases, drive may perform it as async operation, but still significantly limit the overall write IOPS (i.e-75%).

[0106] Based on the above, it should be appreciated that 512B writes may generate a RMW operation that is sub-optimal from the drive's perspective. Along these lines, an extensive amount of 512B writes may pull the overall performance down due to the drive's mechanism and low performance of 512B-writes.

[0107] As an example, imagine a system that for every 4 KB of data (or several pages), writes 512B of MD. Since the 512B IOPS are very limited, the overall write performance of the system will be limited.

[0108] In accordance with certain embodiments, the goal is to find a way around using the 512B IOs, and sticking as much as possible to IU that is optimal for the drive's performance.

[0109] So, in these embodiments, the goal is to achieve the best write performance possible, while using a commodity SSD as a device for persistent log/journal/write-cache without compromising the latency, IOPS or bandwidth.

[0110] To avoid 512B writes overhead, the application (log/journal/write-cache) which is using such drives is configured to inflate all 512B sectors to 4 KB-meaning the application will pad such sectors with 3,584B of zeros thus achieving a full 4K page write-operation that is aligned with the drive's native-page-size and hence optimal for the drive's performance.

[0111] As an example, if the application required 8 GB, out of them 4 GB of 512B sectors (i.e. 8M sectors)+4 GB of 4k pages (1M pages), it will now allocate 36 GB: 32 GB for the 8M sectors that were inflated to 4k pages, and an additional 4 GB for the 4k pages.

[0112] Note that given the application in discussion is mostly log/journal/write-cache that initially has a relatively small footprint (tens to hundreds of GB) compared to a drive's capacity (4 TB to 16 TB). Accordingly, inflating the footprint is acceptable.

[0113] Doing that solves the problem for low 512B IOPS (and increased latency)—now incoming data can be written at the rate of the drive's 4k IOPS. However, the inflation results with two side effects: [0114] a. Endurance: instead of writing 4.5 KB, application now writes 8 KB. [0115] b. Performance: the overall system IOPS is improved, a new limitation may exist in certain drives such as PCIe devices (say 3.2 GB/s for a PCIe Gen5 drives). For such devices, the cost of inflation may take its toll if performed blindly.

[0116] For example, say we have a drive that uses a Gen3 PCIe x4 link that can provide ~3.2 GB/s. Then for large system-IOs that can saturate the drive's PCIe link, we were previously writing 8.5 KB (8 KB+512B), but after inflation we need to write 12 KB (8 KB +4 KB). This means that our previous max BW (to saturate the drive's link) was 3.01 GB/s (3.2 GB/s x 8/8.5-6.25% BW overhead), and after inflating it, we are down to 2.1 GB/s (3.2 GB/s x 8/12-50% BW overhead).

[0117] To mitigate that large BW overhead, some embodiments involve dynamically modifying the write size according to the ingest system IO size. So, for small blocks writes (i.e. 4K/8K/etc.,) which requires high IOPS for the MD descriptor, but are not limited by PCIe BW, the circuitry writes a padded descriptor using 4K writes (with a significant link overhead), which will maximize our IOPS.

[0118] While for large block writes which require high BW, and are limited by the overhead of the PCIe BW, the circuitry falls back to 512B sector (still written in 4K allocations), to reduce the link overhead. In this case, the circuitry simply lets the drive do its thing (e.g. RMW), thus paying the toll of IOPS overhead which is not the bounding-factor.

[0119] It should be appreciated that, as most endurance in the field is driven by large IOs, refraining from inflating large data will also help reduce the endurance overhead.

[0120] In accordance with certain embodiments, small write IOs involve inflating the MD-descriptors and allocate them in 4k pages (corresponding to the drive native page size). In some situations, the system may be tested to find a threshold of write IO size where below it we write a sector padded to 4K, and above it inflation is disabled/avoided and the data is written as a non-padded 512B sector to a 4K inflated page (obviously all the pages are zeroed at first).

[0121] In some embodiments, the drive's endurance is addressed by using SSDs with healthy capacity (e.g., 4 TB, 8 TB, 16 TB), limiting the size of the write cache (e.g., 4 GB, 8 GB, etc.) and giving back the remaining capacity to the SSD firmware.

[0122] Since these drives would serve as log/journal/write-cache that only require few tens up to hundreds of GBs, compared to the drive's capacity of 4 TB to 16 TB, such configuration allows for short-stroking the drive up to 98%-99% that will result in a write amplification factor (WAF) of 1.0+/- (as most writes would become sequential). [0123] SSD used as a caching device are short-stroked to the size of cache (i.e. extra capacity will not be used for any other purposes) [0124] Such operation allows WPD of the SSD to increase (from 1× to ~5×) [0125] Depending on the WPD requirements of the cache, appropriate size of the SSD can be chosen [0126] For example, if 32 TB of WPD is needed (Page-Blocks+Page-Descriptors) for cache, then 8 TB SSD (which gives ~5× WPD w/short-stroking) can be chosen.

[0127] The individual features of the various embodiments, examples, and implementations disclosed within this document can be combined in any desired manner that makes technological sense. Furthermore, the individual features are hereby combined in this manner to form all possible combinations, permutations and variants except to the extent that such combinations, permutations and/or variants have been explicitly excluded or are impractical. Support for such combinations, permutations and variants is considered to exist within this document. Such modifications and enhancements are intended to belong to various embodiments of the disclosure.

## Claims

1. A method of processing write operation information, the method comprising: receiving, as some of the write operation information, user data and corresponding metadata; based on a size of the user data being less than a threshold, generating inflated data which includes the user data and the corresponding metadata, the inflated data having a size that is larger than a total size of the user data and the corresponding metadata, and the inflated data naturally aligning on storage boundaries of solid state device (SSD) storage; after generating the inflated data, storing the inflated data in the SSD storage; receiving, as additional write operation information, additional user data and additional corresponding metadata; and based on a size of the additional user data being greater than the threshold, refraining from generating additional inflated data which includes the user data and the corresponding metadata, and storing the additional user data and the additional corresponding metadata in the SSD storage.
2. The method of claim 1 wherein the SSD storage forms a write cache of a storage array; and wherein storing the inflated data in the SSD storage includes: writing the inflated data into the write cache of the storage array.
3. The method of claim 2 wherein the user data is application data; wherein the corresponding metadata is constructed and arranged to identify write operation details for storing the application data in non-volatile memory of the storage array, the non-volatile memory being different from the

write cache, the corresponding metadata having a size which is smaller than that of the application data; and wherein generating the inflated data includes: padding the corresponding metadata to form padded metadata having an inflated size that is larger than the size of the corresponding metadata.

**4.** The method of claim 3 wherein writing the inflated data into the write cache includes: writing, as the inflated data, the application data and the padded metadata into the write cache to enable late binding of the application data into the non-volatile memory.

**5.** The method of claim 4, further comprising: after the application data and the padded metadata are written into the write cache, performing a late bind operation which commits the application data to the non-volatile memory of the storage array.

**6.** The method of claim 3, further comprising: prior to receiving the user data and the corresponding metadata, performing a comparison operation which compares a size of the user data to a threshold and provides a comparison result indicating that the size of the user data is lower than the threshold.

**7.** The method of claim 6 wherein padding the corresponding metadata includes: in response to the comparison result indicating that the size of the user data is lower than the threshold, creating a full page which includes the corresponding metadata and zeroes, the full page being aligned with a native page size of the SSD storage.

**8.** The method of claim 7, further comprising: performing an additional comparison operation which compares the size of the additional user data to the threshold and provides an additional comparison result indicating that the size of the additional user data is greater than the threshold.

**9.** The method of claim 3, further comprising: prior to receiving the user data and the corresponding metadata, forming the write cache from a set of SSDs.

**10.** The method of claim 9 wherein the set of SSDs provides X amount of storage, X amount being at least 1 Terabyte; and wherein forming the write cache from the set of SSDs includes: allocating 10% of the X amount of storage to the write cache, and allocating at least 80% of the X amount of storage to a garbage collection service to minimize write amplification resulting from garbage collecting the write cache.

**11.** Storage equipment, comprising: solid state device (SSD) storage; and control circuitry coupled with the SSD storage, the control circuitry being constructed and arranged to perform a method of: receiving, as write operation information, user data and corresponding metadata, based on a size of the user data being less than a threshold, generating inflated data which includes the user data and the corresponding metadata, the inflated data having a size that is larger than a total size of the user data and the corresponding metadata, and the inflated data naturally aligning on storage boundaries of the SSD storage, and after generating the inflated data, storing the inflated data in the SSD storage; receiving, as additional write operation information, additional user data and additional corresponding metadata; and based on a size of the additional user data being greater than the threshold, refraining from generating additional inflated data which includes the user data and the corresponding metadata, and storing the additional user data and the additional corresponding metadata in the SSD storage.

**12.** Storage equipment as in claim 11 wherein the SSD storage forms a write cache; and wherein storing the inflated data in the SSD storage includes: writing the inflated data into the write cache.

**13.** Storage equipment as in claim 12, further comprising: non-volatile memory coupled with the control circuitry; wherein the user data is application data; wherein the corresponding metadata is constructed and arranged to identify write operation details for storing the application data in the non-volatile memory, the corresponding metadata having a size which is smaller than that of the application data; and wherein generating the inflated data includes: padding the corresponding metadata to form padded metadata having an inflated size that is larger than the size of the corresponding metadata.

**14.** A computer program product having a non-transitory computer readable medium which stores a

set of instructions to process write operation information; the set of instructions, when carried out by computerized circuitry, causing the computerized circuitry to perform a method of: receiving, as some of the write operation information, user data and corresponding metadata; based on a size of the user data being less than a threshold, generating inflated data which includes the user data and the corresponding metadata, the inflated data having a size that is larger than a total size of the user data and the corresponding metadata, and the inflated data naturally aligning on storage boundaries of solid state device (SSD) storage; after generating the inflated data, storing the inflated data in the SSD storage; receiving, as additional write operation information, additional user data and additional corresponding metadata; and based on a size of the additional user data being greater than the threshold, refraining from generating additional inflated data which includes the user data and the corresponding metadata, and storing the additional user data and the additional corresponding metadata in the SSD storage.

**15.** The computer program product of claim 14 wherein the SSD storage forms a write cache of a storage array; and wherein storing the inflated data in the SSD storage includes: writing the inflated data into the write cache of the storage array.

**16.** The computer program product of claim 15 wherein the user data is application data; wherein the corresponding metadata is constructed and arranged to identify write operation details for storing the application data in non-volatile memory of the storage array, the non-volatile memory being different from the write cache, the corresponding metadata having a size which is smaller than that of the application data; and wherein generating the inflated data includes: padding the corresponding metadata to form padded metadata having an inflated size that is larger than the size of the corresponding metadata.

**17.** The method of claim 1 wherein the SSD storage includes a set of SSDs which forms a write cache and secondary storage; wherein storing the inflated data in the SSD storage includes: writing the inflated data which includes the user data and the corresponding metadata into the write cache; and wherein the method further comprises: after writing the inflated data into the write cache, performing a commit operation which synchronizes the user data into the secondary storage.

**18.** The method of claim 17 wherein performing the commit operation includes: saving the user data in the secondary storage based on the corresponding metadata.

**19.** Storage equipment as in claim 11 wherein the SSD storage includes a set of SSDs which forms a write cache and secondary storage; wherein storing the inflated data in the SSD storage includes: writing the inflated data which includes the user data and the corresponding metadata into the write cache; and wherein the method further comprises: after writing the inflated data into the write cache, performing a commit operation which synchronizes the user data into the secondary storage.

**20.** The computer program product of claim 14 wherein the SSD storage includes a set of SSDs which forms a write cache and secondary storage; wherein storing the inflated data in the SSD storage includes: writing the inflated data which includes the user data and the corresponding metadata into the write cache; and wherein the method further comprises: after writing the inflated data into the write cache, performing a commit operation which synchronizes the user data into the secondary storage.

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